T3 A_+0: general EFT1 RGEs

Matchete/RGBeta symbolic output in mathematical notation

A_+0

Additional quartics: 5, additional bilinears: 1, extra Yukawa: y1Conj.
General matching and Wilson coefficients are not computed in this symbolic report.

One-loop renormalisable beta functions

$16\pi^2\, dX/d\ln\mu = \beta_X$

g_1

$$16\pi^2 \frac{dg_1}{d\ln\mu} = \frac{47{g_1}^3}{6}$$

g_2

$$16\pi^2 \frac{dg_2}{d\ln\mu} = -\frac{5{g_2}^3}{2}$$

g_3

$$16\pi^2 \frac{dg_3}{d\ln\mu} = -7{g_3}^3$$

Y_u

$$\begin{aligned}
16\pi^2 \frac{dY_u}{d\ln\mu} = {} & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_u)_{ij} \\
& + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_u)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_u)_{ij} \\
& - \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_u\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_u\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_u)_{ij} \\
& - 8{g_3}^2\,(Y_u)_{ij} \\
& - \frac{17}{12}{g_1}^2\,(Y_u)_{ij}
\end{aligned}$$

Y_d

$$\begin{aligned}
16\pi^2 \frac{dY_d}{d\ln\mu} = {} & \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_d)_{ij} \\
& + 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_d)_{ij} \\
& + \frac{3}{2}(\Big(Y_d\ (Y_d)^\dagger\ Y_d\Big)_{ij} - \Big(Y_u\ (Y_u)^\dagger\ Y_d\Big)_{ij}) \\
& - \frac{9}{4}{g_2}^2\,(Y_d)_{ij} \\
& - 8{g_3}^2\,(Y_d)_{ij} \\
& - \frac{5}{12}{g_1}^2\,(Y_d)_{ij}
\end{aligned}$$

Y_e

$$\begin{aligned}16\pi^2 \frac{dY_e}{d\ln\mu} = & 3\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big)\,(Y_e)_{ij} \\ & + 3\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)\,(Y_e)_{ij} \\ & + \mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big)\,(Y_e)_{ij} \\ & + \frac{3}{2}\Big(Y_e\ (Y_e)^\dagger\ Y_e\Big)_{ij} \\ & - \frac{9}{4}{g_2}^2\,(Y_e)_{ij} \\ & - \frac{15}{4}{g_1}^2\,(Y_e)_{ij}\end{aligned}$$

m_1^2

$$\begin{aligned}16\pi^2 \frac{dm_1^2}{d\ln\mu} = & 12\mathrm{muT3GeneralS1Sq}\,(\mathrm{lambdaT3GeneralQ001300I1})^* \\ & + 12\mathrm{lambdaT3GeneralQ001300I1}\,(\mathrm{muT3GeneralS1Sq})^* \\ & + 6\lambda_{12}^{(1)} m_2^2 \\ & + 4\lambda_{S_1}^{(1)} m_1^2\end{aligned}$$

m_2^2

$$\begin{aligned}16\pi^2 \frac{dm_2^2}{d\ln\mu} = & 4\mathrm{muT3GeneralS1Sq}\,(\mathrm{lambdaT3GeneralQ000211I1})^* \\ & + 4\mathrm{lambdaT3GeneralQ000211I1}\,(\mathrm{muT3GeneralS1Sq})^* \\ & - 12{g_2}^2 m_2^2 \\ & - 6{g_1}^2 m_2^2 \\ & + 2\lambda_{12}^{(1)} m_1^2 \\ & + m_2^2(8\lambda_{S_2}^{(1)} + 4\lambda_{S_2}^{(A)})\end{aligned}$$

λ_1

$$\begin{aligned}16\pi^2 \frac{d\lambda_1}{d\ln\mu} = & \lambda_1(12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\ & - 12\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\ Y_d\ (Y_d)^\dagger\Big) \\ & - 4\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\ Y_e\ (Y_e)^\dagger\Big) \\ & - 12\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\ Y_u\ (Y_u)^\dagger\Big) \\ & + 4\lambda_5\,(\lambda_5)^* \\ & + 4\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^* \\ & + 8\mathrm{lambdaT3GeneralQ110200I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^* \\ & + \frac{9{g_2}^4}{4} \\ & + {g_1}^2\left(\frac{3{g_2}^2}{2} - 3\lambda_1\right) \\ & - 9{g_2}^2\lambda_1\end{aligned}$$

$$
\begin{aligned}
&+ \frac{3{g_1}^4}{4} \\
&+ 12{\lambda_1}^2 \\
&+ 2{\lambda_{HS_1}^{(1)}}^2 \\
&+ 6{\lambda_{HS_2}^{(1)}}^2 \\
&+ 3{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{S_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_1}^{(1)}}{d\ln\mu} = &\ 12\text{lambdaT3GeneralQ000211I1} \left(\text{lambdaT3GeneralQ000211I1}\right)^* \\
&+ 288\text{lambdaT3GeneralQ000400I1} \left(\text{lambdaT3GeneralQ000400I1}\right)^* \\
&+ 108\text{lambdaT3GeneralQ001300I1} \left(\text{lambdaT3GeneralQ001300I1}\right)^* \\
&+ 8\text{lambdaT3GeneralQ110200I1} \left(\text{lambdaT3GeneralQ110200I1}\right)^* \\
&+ 6\lambda_{12}^{(1)2} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 10\lambda_{S_1}^{(1)2}
\end{aligned}
$$

$\lambda_{S_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(1)}}{d\ln\mu} = &\ 8\text{lambdaT3GeneralQ000211I1} \left(\text{lambdaT3GeneralQ000211I1}\right)^* \\
&+ 22{g_2}^4 \\
&+ {g_1}^2 \left(8{g_2}^2 - 12\lambda_{S_2}^{(1)}\right) \\
&- 24{g_2}^2\lambda_{S_2}^{(1)} \\
&+ 12{g_1}^4 \\
&+ 2\lambda_{12}^{(1)2} \\
&+ 4{\lambda_{HS_2}^{(1)}}^2 \\
&+ 14\lambda_{S_2}^{(1)2} \\
&+ 8\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 16\lambda_{S_2}^{(A)2}
\end{aligned}
$$

$\lambda_{HS_1}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_1}^{(1)}}{d\ln\mu} = &\ \lambda_{HS_1}^{(1)}(6\,\mathrm{Tr}\left(Y_d\ \left(Y_d\right)^\dagger\right) + 2\,\mathrm{Tr}\left(Y_e\ \left(Y_e\right)^\dagger\right) + 6\,\mathrm{Tr}\left(Y_u\ \left(Y_u\right)^\dagger\right)) \\
&+ 12\lambda_5\ \left(\lambda_5\right)^* \\
&+ 12\text{lambdaT3GeneralQ001300I1} \left(\text{lambdaT3GeneralQ110200I1}\right)^* \\
&+ 4\text{lambdaT3GeneralQ110200I1}(3\left(\text{lambdaT3GeneralQ001300I1}\right)^* + 4\left(\text{lambdaT3GeneralQ110200I1}\right)^*) \\
&+ 12\text{lambdaT3GeneralQ021010I1} \left(\text{lambdaT3GeneralQ021010I1}\right)^* \\
&- \frac{9{g_2}^2\lambda_{HS_1}^{(1)}}{2}
\end{aligned}
$$

$$
\begin{aligned}
&- \frac{3{g_1}^2\lambda_{HS_1}^{(1)}}{2} \\
&+ 6\lambda_{12}^{(1)}\lambda_{HS_2}^{(1)} \\
&+ 6\lambda_1\lambda_{HS_1}^{(1)} \\
&+ 4{\lambda_{HS_1}^{(1)}}^2 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_{S_1}^{(1)}
\end{aligned}
$$

$\lambda_{HS_2}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(1)}}{d\ln\mu} = {} & \lambda_{HS_2}^{(1)}(6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{S_2}^{(A)}) \\
& + 4\lambda_5\,(\lambda_5)^* \\
& + 4\mathrm{lambdaT3GeneralQ110200I1}\,(\mathrm{lambdaT3GeneralQ000211I1})^* \\
& + 4\mathrm{lambdaT3GeneralQ000211I1}\,(\mathrm{lambdaT3GeneralQ110200I1})^* \\
& + 4\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^* \\
& + 6{g_2}^4 \\
& - \frac{33{g_2}^2\lambda_{HS_2}^{(1)}}{2} \\
& + 3{g_1}^4 \\
& - \frac{15{g_1}^2\lambda_{HS_2}^{(1)}}{2} \\
& + 2\lambda_{12}^{(1)}\lambda_{HS_1}^{(1)} \\
& + 6\lambda_1\lambda_{HS_2}^{(1)} \\
& + 4{\lambda_{HS_2}^{(1)}}^2 \\
& + 8\lambda_{HS_2}^{(1)}\lambda_{S_2}^{(1)} \\
& + 6{\lambda_{HS_2}^{(A)}}^2
\end{aligned}
$$

$\lambda_{12}^{(1)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{12}^{(1)}}{d\ln\mu} = {} & 4\lambda_5\,(\lambda_5)^* \\
& + 12\mathrm{lambdaT3GeneralQ001300I1}\,(\mathrm{lambdaT3GeneralQ000211I1})^* \\
& + 4\mathrm{lambdaT3GeneralQ000211I1}(4\,(\mathrm{lambdaT3GeneralQ000211I1})^* + 3\,(\mathrm{lambdaT3GeneralQ001300I1})^*) \\
& + 4\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^* \\
& - 12{g_2}^2\lambda_{12}^{(1)} \\
& - 6{g_1}^2\lambda_{12}^{(1)} \\
& + 4{\lambda_{12}^{(1)}}^2 \\
& + 4\lambda_{12}^{(1)}\lambda_{S_1}^{(1)} \\
& + 8\lambda_{12}^{(1)}\lambda_{S_2}^{(1)} \\
& + 4\lambda_{12}^{(1)}\lambda_{S_2}^{(A)} \\
& + 4\lambda_{HS_1}^{(1)}\lambda_{HS_2}^{(1)}
\end{aligned}
$$

λ_5

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_5}{d\ln\mu} = \lambda_5(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 4\lambda_{HS_2}^{(A)}) \\
&+ 4\,(\mathrm{lambdaT3GeneralQ021010I1})^*\,((\mathrm{lambdaT3GeneralQ000211I1})^* + 2\,(\mathrm{lambdaT3GeneralQ110200I1})^*) \\
&- \frac{21{g_2}^2\lambda_5}{2} \\
&- \frac{9{g_1}^2\lambda_5}{2} \\
&+ 2\lambda_{12}^{(1)}\lambda_5 \\
&+ 2\lambda_1\lambda_5 \\
&+ 4\lambda_{HS_1}^{(1)}\lambda_5 \\
&+ 4\lambda_{HS_2}^{(1)}\lambda_5
\end{aligned}
$$

$\lambda_{HS_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{HS_2}^{(A)}}{d\ln\mu} = \lambda_{HS_2}^{(A)}(&6\,\mathrm{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\mathrm{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\mathrm{Tr}\Big(Y_u\ (Y_u)^\dagger\Big) + 16\lambda_{S_2}^{(A)}) \\
&+ \frac{8}{3}\lambda_5\,(\lambda_5)^* \\
&+ \frac{8}{3}\mathrm{lambdaT3GeneralQ021010I1}\,(\mathrm{lambdaT3GeneralQ021010I1})^* \\
&+ {g_1}^2\left(4{g_2}^2 - \frac{15\lambda_{HS_2}^{(A)}}{2}\right) \\
&- \frac{33{g_2}^2\lambda_{HS_2}^{(A)}}{2} \\
&+ 2\lambda_1\lambda_{HS_2}^{(A)} \\
&+ 8\lambda_{HS_2}^{(1)}\lambda_{HS_2}^{(A)} \\
&+ 2\lambda_{HS_2}^{(A)}\lambda_{S_2}^{(1)}
\end{aligned}
$$

$\lambda_{S_2}^{(A)}$

$$
\begin{aligned}
16\pi^2 \frac{d\lambda_{S_2}^{(A)}}{d\ln\mu} = &-2{g_2}^4 \\
&+ {g_1}^2\left(8{g_2}^2 - 12\lambda_{S_2}^{(A)}\right) \\
&- 24{g_2}^2\lambda_{S_2}^{(A)} \\
&+ 3{\lambda_{HS_2}^{(A)}}^2 \\
&+ 12\lambda_{S_2}^{(1)}\lambda_{S_2}^{(A)} \\
&+ 20{\lambda_{S_2}^{(A)}}^2
\end{aligned}
$$

$lambdaT3GeneralQ000211I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000211I1}{d\ln\mu} = &\,4\,(\lambda_5)^*\,(\mathrm{lambdaT3GeneralQ021010I1})^* \\
&+ 24\mathrm{lambdaT3GeneralQ000400I1}\,(\mathrm{lambdaT3GeneralQ000211I1})^*
\end{aligned}
$$

$$
\begin{aligned}
&- 12{g_2}^2\text{lambdaT3GeneralQ000211I1} \\
&- 6{g_1}^2\text{lambdaT3GeneralQ000211I1} \\
&+ \lambda_{12}^{(1)}(8\text{lambdaT3GeneralQ000211I1} + 6\text{lambdaT3GeneralQ001300I1}) \\
&+ 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ110200I1} \\
&+ 2\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ000211I1} \\
&+ 8\lambda_{S_2}^{(1)}\text{lambdaT3GeneralQ000211I1} \\
&+ 4\lambda_{S_2}^{(A)}\text{lambdaT3GeneralQ000211I1}
\end{aligned}
$$

$lambdaT3GeneralQ000400I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ000400I1}{d\ln\mu} = &12\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ000400I1} \\
&+ 3\text{lambdaT3GeneralQ000211I1}^2 \\
&+ 9\text{lambdaT3GeneralQ001300I1}^2 \\
&+ 2\text{lambdaT3GeneralQ110200I1}^2
\end{aligned}
$$

$lambdaT3GeneralQ001300I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ001300I1}{d\ln\mu} = &72\text{lambdaT3GeneralQ000400I1}\,(\text{lambdaT3GeneralQ001300I1})^* \\
&+ 6\lambda_{12}^{(1)}\text{lambdaT3GeneralQ000211I1} \\
&+ 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ110200I1} \\
&+ 18\lambda_{S_1}^{(1)}\text{lambdaT3GeneralQ001300I1}
\end{aligned}
$$

$lambdaT3GeneralQ021010I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ021010I1}{d\ln\mu} = &\text{lambdaT3GeneralQ021010I1}(6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ 4\,(\lambda_5)^*\,((\text{lambdaT3GeneralQ000211I1})^* + 2\,(\text{lambdaT3GeneralQ110200I1})^*) \\
&- \frac{21{g_2}^2\text{lambdaT3GeneralQ021010I1}}{2} \\
&- \frac{9{g_1}^2\text{lambdaT3GeneralQ021010I1}}{2} \\
&+ 2\lambda_{12}^{(1)}\text{lambdaT3GeneralQ021010I1} \\
&+ 2\lambda_1\text{lambdaT3GeneralQ021010I1} \\
&+ 4\lambda_{HS_1}^{(1)}\text{lambdaT3GeneralQ021010I1} \\
&+ 4\lambda_{HS_2}^{(1)}\text{lambdaT3GeneralQ021010I1} \\
&+ 4\lambda_{HS_2}^{(A)}\text{lambdaT3GeneralQ021010I1}
\end{aligned}
$$

$lambdaT3GeneralQ110200I1$

$$
\begin{aligned}
16\pi^2 \frac{dlambdaT3GeneralQ110200I1}{d\ln\mu} = &\text{lambdaT3GeneralQ110200I1}(6\,\text{Tr}\Big(Y_d\ (Y_d)^\dagger\Big) + 2\,\text{Tr}\Big(Y_e\ (Y_e)^\dagger\Big) + 6\,\text{Tr}\Big(Y_u\ (Y_u)^\dagger\Big)) \\
&+ 12\,(\lambda_5)^*\,(\text{lambdaT3GeneralQ021010I1})^*
\end{aligned}
$$

$$
\begin{aligned}
&+ 24\text{lambdaT3GeneralQ000400I1}\,(\text{lambdaT3GeneralQ110200I1})^{*} \\
&- \frac{9{g_2}^2\text{lambdaT3GeneralQ110200I1}}{2} \\
&- \frac{3{g_1}^2\text{lambdaT3GeneralQ110200I1}}{2} \\
&+ 6\lambda_1\text{lambdaT3GeneralQ110200I1} \\
&+ \lambda^{(1)}_{HS_1}(6\text{lambdaT3GeneralQ001300I1} + 8\text{lambdaT3GeneralQ110200I1}) \\
&+ 6\lambda^{(1)}_{HS_2}\text{lambdaT3GeneralQ000211I1} \\
&+ 2\lambda^{(1)}_{S_1}\text{lambdaT3GeneralQ110200I1}
\end{aligned}
$$

$muT3GeneralS1Sq$

$$
\begin{aligned}
16\pi^2\,\frac{dmuT3GeneralS1Sq}{d\ln\mu} &= 24\text{lambdaT3GeneralQ000400I1}\,(\text{muT3GeneralS1Sq})^{*} \\
&\quad + 2\lambda^{(1)}_{S_1}\text{muT3GeneralS1Sq} \\
&\quad + 6\text{lambdaT3GeneralQ000211I1}m_2^2 \\
&\quad + 6\text{lambdaT3GeneralQ001300I1}m_1^2
\end{aligned}
$$